

TABLE II
WINDOWED OPERAND DEMANDS OF REPRESENTATIVE LLM WORKLOADS UNDER EXPLICIT CIM MAPPINGS

Workload / operator	Window and residency mapping	Key dimensions / parameters	Q_R (MiB)	Q_S (MiB)	RI
S1 K projection	G0: pre-resident; decode	$U = 1; W_K : 512 \times 3584$	0	0.0068	∞
S2 K projection	G1: load once; decode	$B = U = 1$	3.5	0.0068	$\frac{1}{512}$
S3 K, 7B / 14B	G1: load once; decode	$B = U = 8$; shared W_K	3.5 / 10	0.0547 / 0.0781	$\frac{1}{64} / \frac{1}{128}$
S4 MLP gate	G1: load once; prefill	$U = 2048; W_g : 18944 \times 3584$	129.5	14	$\frac{4}{37}$
S5 MLP gate	G2: 4 placement epochs	$U = 2048$; 512 vectors/epoch	518	14	$\frac{1}{37}$
S6 Causal prefill	A1: empty KV; append prefixes	$B = 1, C = 0, N = 2048$	4	126.0547	$\frac{16135}{512}$
S7 KV-append decode	A1: keep history; append one	$B = 8, N = 1, L_v = 8192$	0.0156	3.5547	227.5
S8 Gate input grad. dX	T1: pre-resident $W^\top$	$dY : 2048 \times 18944$	0	74	∞
S9 Gate weight grad. dW	T1: load $X^\top$ once	$X^\top : 3584 \times 2048$	14	74	$\frac{37}{7}$
S10 Gate update cycle	T1: two weight layouts updated	$U = 2048; A = 1 / 16$	273 / 483	162 / 2592	$\frac{54}{91} / \frac{864}{161}$

All rows use `logical_operator` boundaries and BFI6 (2 Byte/element). One projection or one attention layer is counted; 7B is used except for the paired 7B/14B values in S3. Model dimensions follow fixed official configurations and implementation [1]–[4]. G0/G1/G2: independent weight-stationary projections; $Q_S = U n_{in} b_S$, $Q_R = E n_{out} n_{in} b_R$, with $E = 0, 1, 4$ respectively. A1: $H_q = 28, H_{kv} = 4, d = 128$; one shared KV copy per group, separate K and V layouts. $L_v = C + N$ is the final visible length, including the current token. For $T = NC + N(N + 1)/2$, $Q_S = BH_q(Ndb_Q + Tb_A)$ and $Q_R = BH_{kv}Nd(b_K + b_V)$; QK and AV are separate services [5], [6]. T1: forward, $dX = dYW$, $dW = dY^\top X$; temporary resident $X^\top$ is replaced each microbatch, and pre-resident $W, W^\top$ are each updated once per cycle [7]. S8–S9 are per microbatch; S10 gives cycle totals for $A = 1/16$. S6–S7 and S10 are demand sums with retained components, without a common service-rate claim. The parameterized tile-port export uses arbitrary (r, c) and counts each receiving output tile; ISAAC supports the underlying resident-MVM and partitioned-input concept [8], while these schedules are our reference choices. Payloads are rounded to 4 decimal places where needed (1 MiB=2²⁰ Byte); RI fractions are exact. Softmax, RoPE, bias and other non-matrix work are outside scope. An AV input does not require external materialization of the entire attention matrix; these demands are not FlashAttention HBM traffic [9].

REFERENCES

- [1] Qwen Team, “Qwen2.5-7B: Official configuration and model card,” 2024, revision d149729398750b98c0af14eb82c78cfe92750796. Accessed Sep. 7, 2026. [Online]. Available: <https://huggingface.co/Qwen/Qwen2.5-7B/tree/d149729398750b98c0af14eb82c78cfe92750796>
- [2] —, “Qwen2.5-14B: Official configuration and model card,” 2024, revision 97e1e76335b7017d8f67c08a19d103c0504298c9. Accessed Sep. 7, 2026. [Online]. Available: <https://huggingface.co/Qwen/Qwen2.5-14B/tree/97e1e76335b7017d8f67c08a19d103c0504298c9>
- [3] Hugging Face, “Transformers: Qwen2 model implementation,” 2024, v4.45.2, commit 53fad641cfdb5105e2470bcf3ef17ea8e25cc300, modeling_qwen2.py, lines 265–391, 1089. Accessed Sep. 7, 2026. [Online]. Available: <https://github.com/huggingface/transformers/tree/53fad641cfdb5105e2470bcf3ef17ea8e25cc300/src/transformers/models/qwen2>
- [4] Qwen Team, “Qwen2.5 technical report,” 2024, arXiv:2412.15115v1, Sec. 2 and Table 1. [Online]. Available: <https://arxiv.org/abs/2412.15115v1>
- [5] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, sec. 3.2, Eq. (1); accessed version arXiv:1706.03762v7. [Online]. Available: <https://arxiv.org/abs/1706.03762v7>
- [6] J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zemlyanskiy, F. Lebron, and S. Sanghai, “GQA: Training generalized multi-query transformer models from multi-head checkpoints,” in *Proc. EMNLP*, 2023, pp. 4895–4901, sec. 2.2 and Fig. 2; accessed version arXiv:2305.13245v3. [Online]. Available: <https://aclanthology.org/2023.emnlp-main.298/>
- [7] PyTorch Contributors, “PyTorch: Linear and matrix-multiplication autograd implementations,” 2024, v2.4.1, commit ee1b6804381c57161c477caa380a840a84167676; Linear.cpp, derivatives.yaml, FunctionsManual.cpp (lines 1415–1451). Accessed Sep. 7, 2026. [Online]. Available: <https://github.com/pytorch/pytorch/tree/ee1b6804381c57161c477caa380a840a84167676>
- [8] A. Shafiee, A. Nag, N. Muralimanohar, R. Balasubramonian, J. P. Strachan, M. Hu, R. S. Williams, and V. Srikumar, “ISAAC: A convolutional neural network accelerator with in-situ analog arithmetic in crossbars,” in *Proc. ACM/IEEE ISCA*, 2016, pp. 14–26, author manuscript, Fig. 1 and Secs. III, VI. [Online]. Available: <https://svivek.com/research/publications/isca16.pdf>
- [9] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré, “FlashAttention: Fast and memory-efficient exact attention with IO-awareness,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022, algorithm 1, Sec. 3; accessed version arXiv:2205.14135v2. [Online]. Available: <https://arxiv.org/abs/2205.14135v2>